Introduction to Robotics: Homework I

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Introduction

The homework involves determining the transformation matrices associated with a robotic manipulation task. Specifically, you are required to identify the transformation matrices for all objects in the environment, the relative transformation matrix between the manipulator's end-effector and the object being manipulated, and the transformation matrix for the end-effector that enables the manipulated object to be precisely inserted into a designated hole.

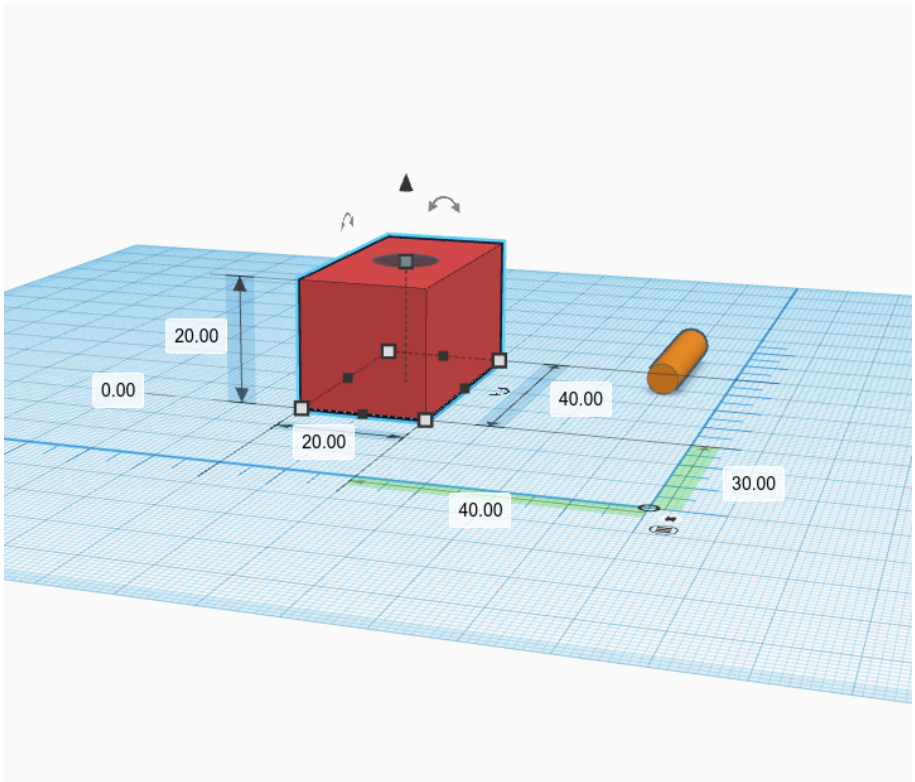

Figure 1: Visualization of the manipulation scenario. **Measurements are in cm.**

1 Description

The setup includes a Franka Panda manipulator equipped with a gripper (Fig. 2), a box featuring a cylindrical hole at its center (represented as the red box in Fig. 1), and a cylinder (shown as the orange cylinder in Fig. 1) that the manipulator must handle and insert into the hole. The Franka Panda is a versatile, 7-DOF robotic arm designed for precise manipulation tasks, widely used in research and industrial applications due to its sensitivity and ease of control. The manipulator's workspace is shown in Fig. 3, 4.

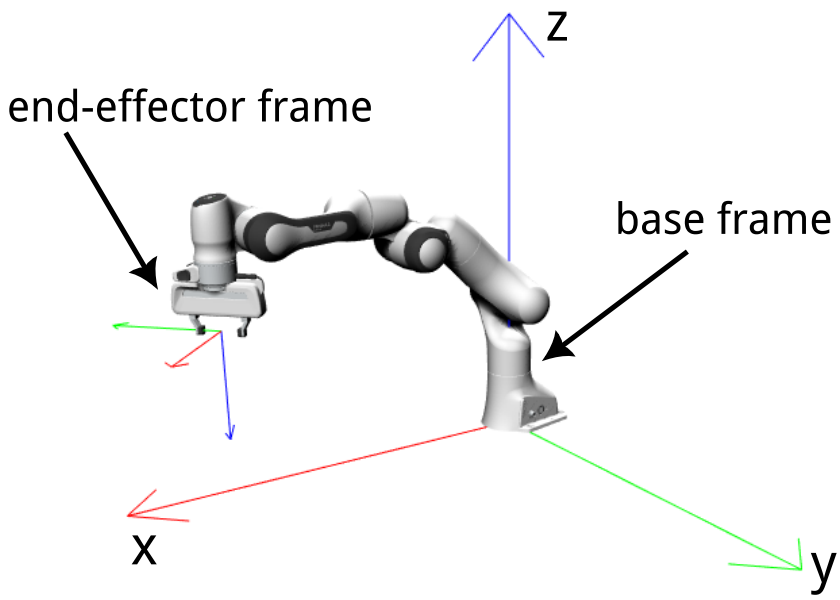

Figure 2: Visualization of the Franka Panda manipulator.

2 Task

Given Fig. 5, Fig. 6, Fig. 7, Fig. 1, Fig. 8, Fig. 9, Fig. 2, Fig. 3, and Fig. 4:

1. Choose the location and orientation of the world frame;
2. Assign a fixed body frame to each object in the scenario;
3. Calculate the transformation matrix for each object relative to the world frame;
4. Select an appropriate transformation matrix for the base frame of the manipulator with respect to the world frame, ensuring it can perform the manipulation task;
5. Determine the relative transformation matrix between the manipulator's end-effector and the cylinder, allowing the manipulator to grasp the cylinder (aka, what should be the orientation and position of the cylinder with respect to the end-effector such that it can be grasped);
6. Calculate the world frame transformation matrix of the cylinder such that, when released, it will be positioned to drop precisely into the hole in the box;
7. Visualize the results using the programming language of your choice.

3 Deliverables

- Detailed report;
- Code for visualization and containing the transformation matrices.

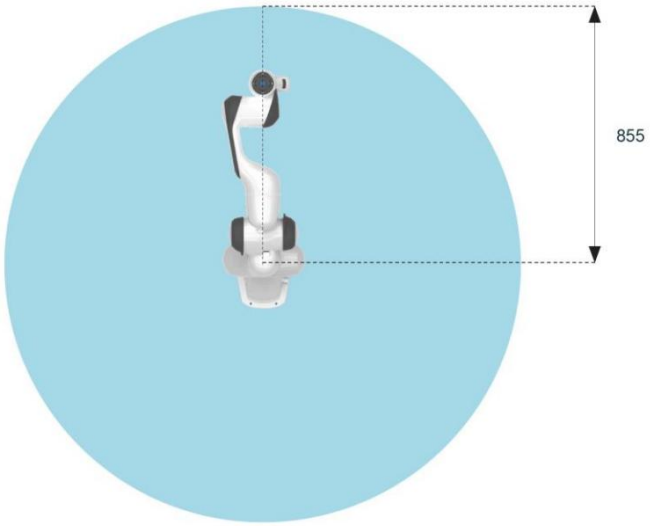

Figure 3: Franka Panda manipulator workspace (top view). **Measurements are in mm.**

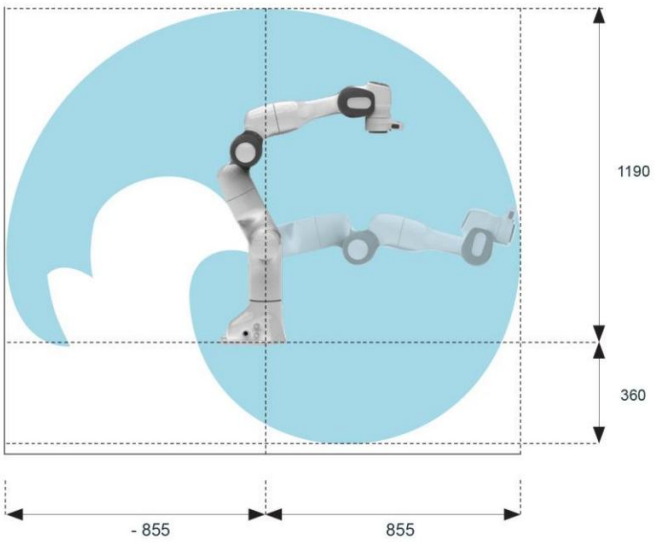

Figure 4: Franka Panda manipulator workspace (side view). **Measurements are in mm.**

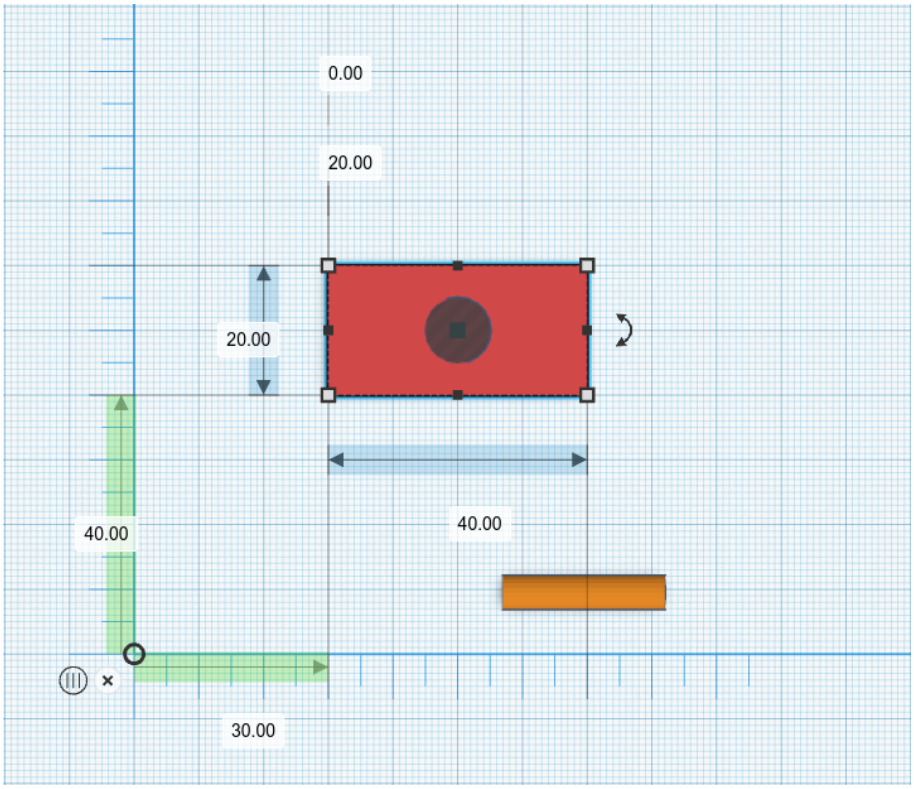

Figure 5: Scenario measurements. **Measurements are in cm.**

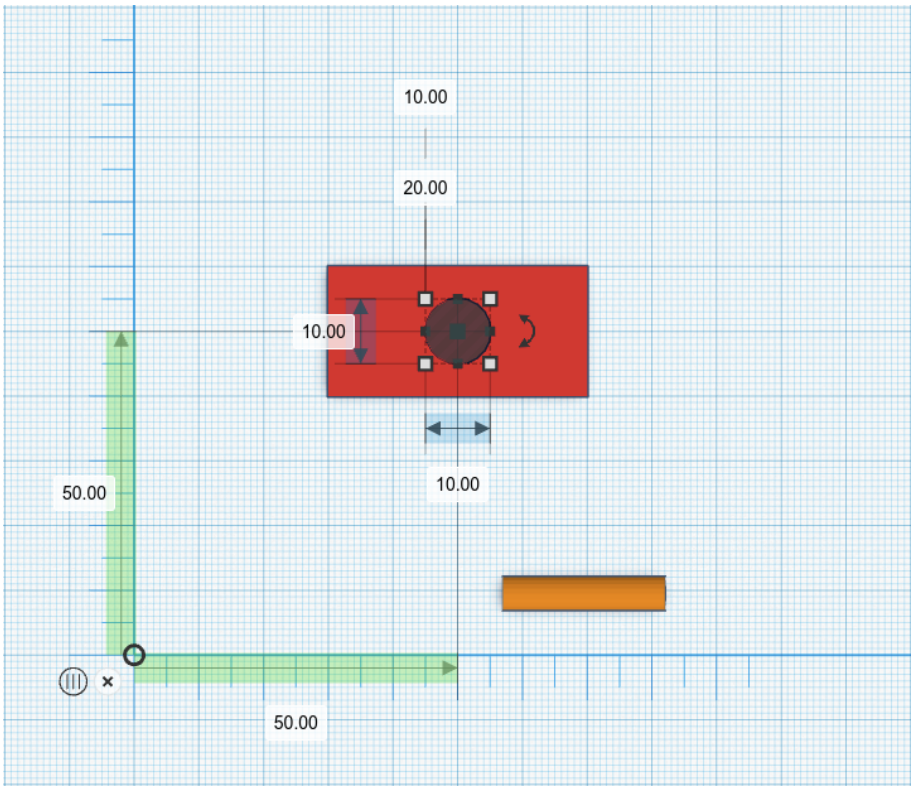

Figure 6: Scenario measurements. **Measurements are in cm.**

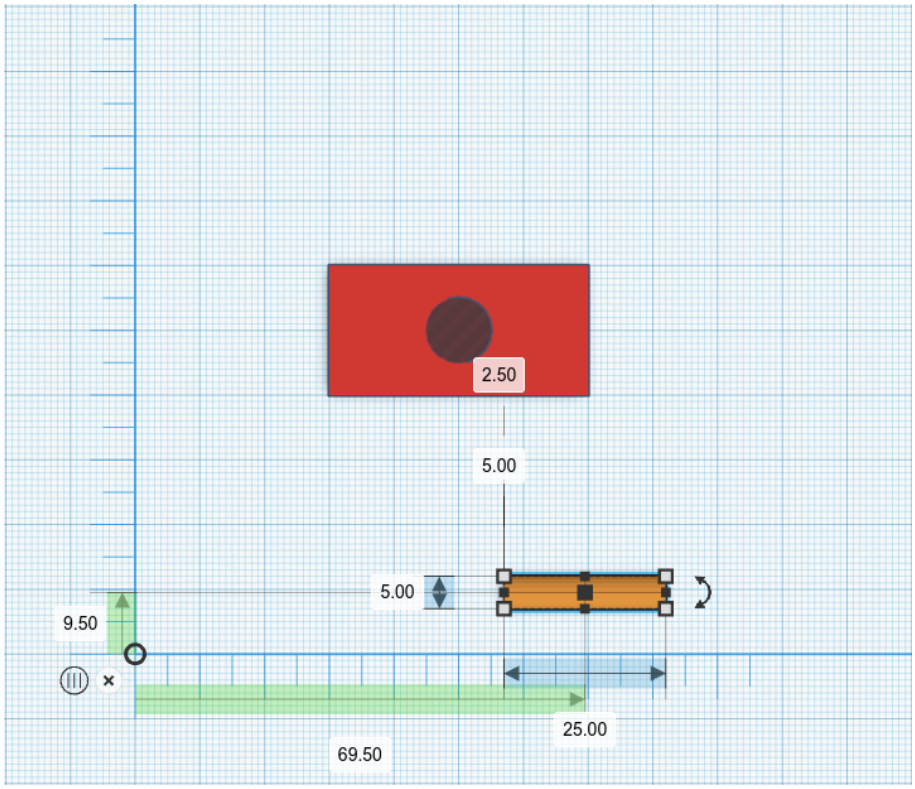

Figure 7: Scenario measurements. **Measurements are in cm.**

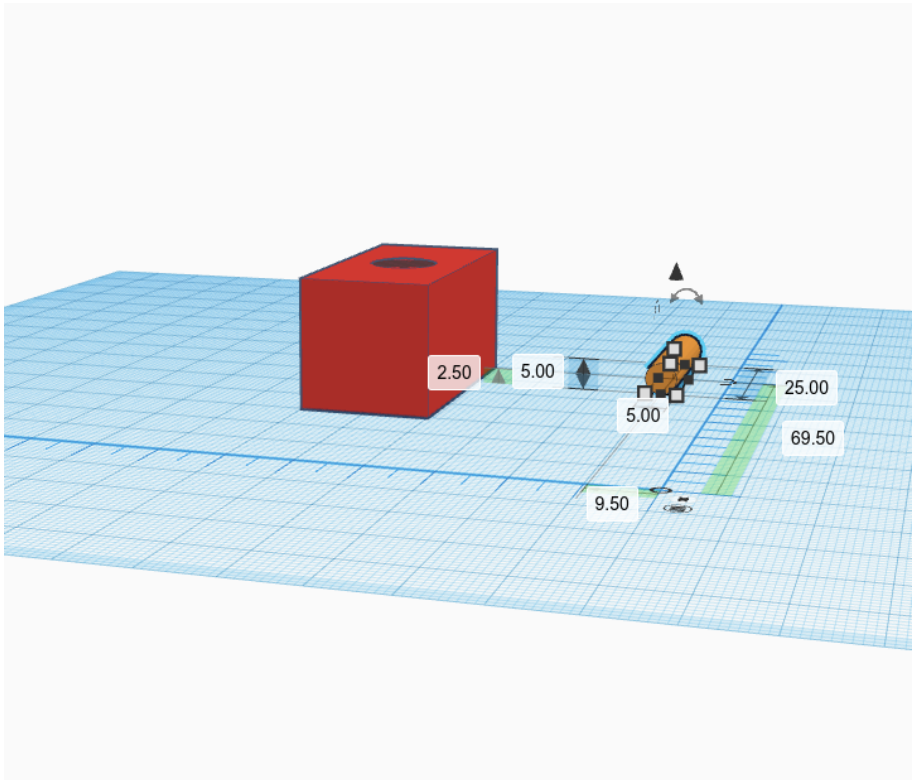

Figure 8: Scenario measurements. **Measurements are in cm.**

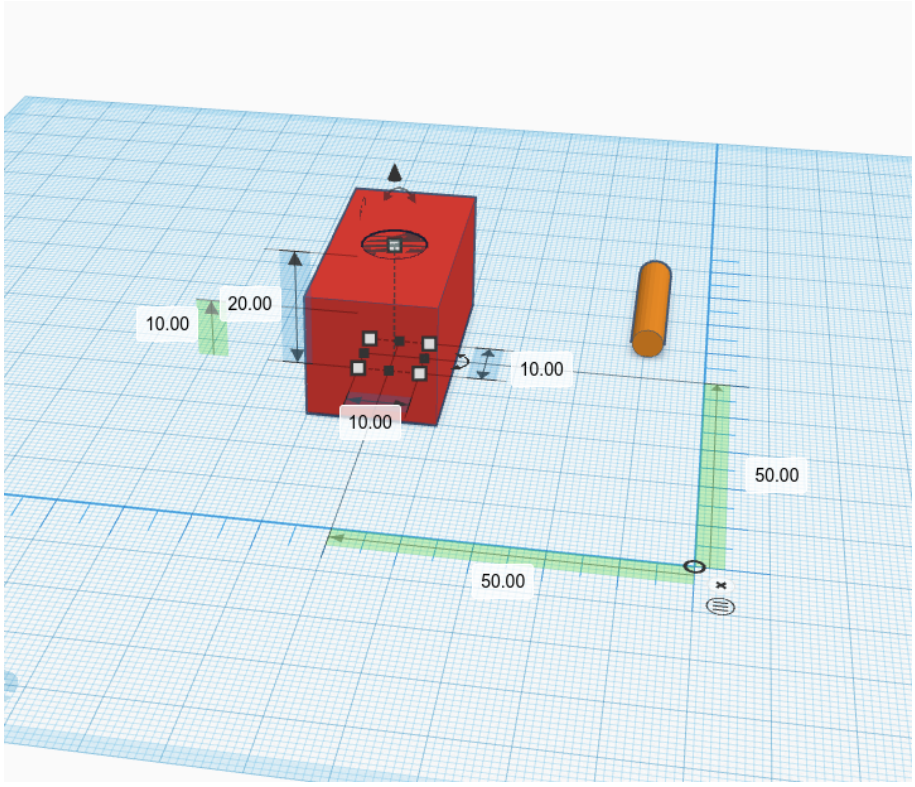

Figure 9: Scenario measurements. **Measurements are in cm.**